

PROGRAMMING FUNDAMENTALS

Spring 2024

27th March

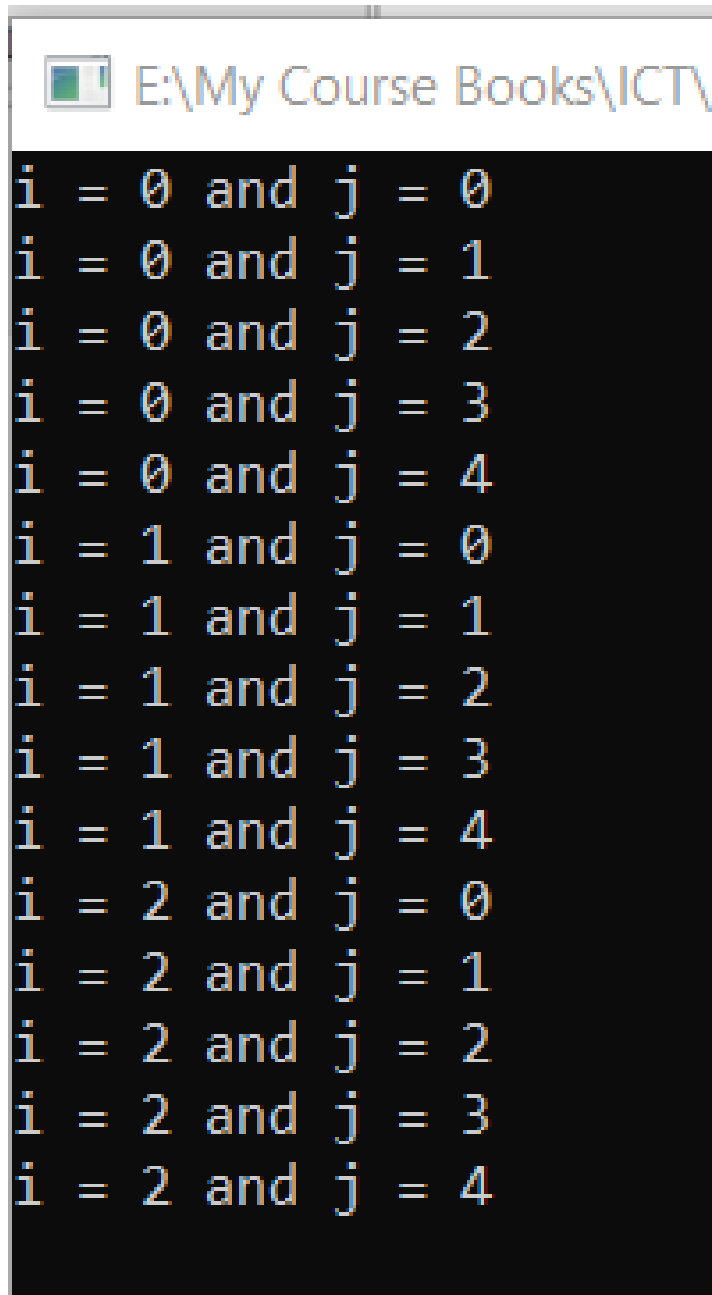
NESTED LOOPS.. Let's practice dry run

Let's dry run

```
#include <iostream>
using namespace std;
int main ()
{
    int i = 0;

    while(i < 3)
    {
        int j = 0;
        while(j < 5)
        {
            cout << "i = " << i << " and j = " << j << endl;
            j++;
        }
        i++;
    }
    return 0;
}
```

Running it ...



A screenshot of a text editor window with the title bar "E:\My Course Books\ICT\". The editor displays a list of 15 lines, each representing a state of two nested loops. The first loop variable 'i' ranges from 0 to 2, and the second loop variable 'j' ranges from 0 to 4. The lines are as follows:

```
i = 0 and j = 0  
i = 0 and j = 1  
i = 0 and j = 2  
i = 0 and j = 3  
i = 0 and j = 4  
i = 1 and j = 0  
i = 1 and j = 1  
i = 1 and j = 2  
i = 1 and j = 3  
i = 1 and j = 4  
i = 2 and j = 0  
i = 2 and j = 1  
i = 2 and j = 2  
i = 2 and j = 3  
i = 2 and j = 4
```

Lets dry run it with input of 3

```
#include <iostream>
using namespace std;
int main()
{
    int size;
    cout << "\n\n Print a pattern like square with # character:\n";
    cout << "-----\n";
    cout << " Input the number of characters for a side: ";
    cin >> size;
    for (int row = 1; row <= size; ++row)
    {
        for (int col = 1; col <= size; ++col)
        {
            cout << "# ";
        }
        cout << endl;
    }
    return 0;
}
```

With input of 3

```
Print a pattern like square with # character:
```

```
-----  
Input the number of characters for a side: 3
```

```
# # #  
# # #  
# # #
```

Changing the inner loop

```
#include <iostream>
using namespace std;
int main()
{
    int size;
    cout << "\n\n Print a pattern like square with # character:\n";
    cout << "-----\n";
    cout << " Input the number of characters for a side: ";
    cin >> size;
    for (int row = 1; row <= size; ++row)
    {
        for (int col = row+1; col <= size; ++col)
        {
            cout << "# ";
        }
        cout << endl;
    }
    return 0;
}
```

With input of 6

 E:\My Course Books\ICT\C++File-F2023\2024\Lec-11.exe

Print a pattern like square with # character:

Input the number of characters for a side: 6

#

#

#

#

#

Let's check this out..

```
#include <iostream>
using namespace std;
int main ()
{
    int i, j;

    for(i=0; i<=5; i++)
    {

        for(j=0; j <= 5; j++)
        {
            cout << i << j <<" \t";
        }

        cout <<"\n";
    }
    return 0;
}
```

Output..



E:\My Course Books\ICT\C++File-F2023\2024\Lec-11.exe

00	01	02	03	04	05
10	11	12	13	14	15
20	21	22	23	24	25
30	31	32	33	34	35
40	41	42	43	44	45
50	51	52	53	54	55

More practice..

```
#include <iostream>
using namespace std;

int main ()
{
    int i, j;

    for(i=0; i<=5; i++)
    {
        for(j=0; j <= 5; j++)
        {
            cout << i;
        }

        cout << "\n";
    }
    return 0;
}
```

Soln.



D:\Books-ICT\C-Programs\nested_loop6.exe

000000

111111

222222

333333

444444

555555

Process exited after 0.2223 seconds w

Press any key to continue . . .

Try this..

```
#include <iostream>
using namespace std;

int main()
{
    int n, i, j;

    cout << "Enter number of rows: ";
    cin >> n;

    for(i = 1; i <= n; i++)
    {
        //print * equal to row number
        for(j = 1; j <= i; j++)
        {
            cout << "* ";
        }
        cout << "\n";
    }
    return 0;
}
```

Output..

```
D:\Books-ICT\C-Programs\nested_loop7.exe  
Enter number of rows: 20  
*  
* *  
* * *  
* * * *  
* * * * *  
* * * * * *  
* * * * * * *  
* * * * * * * *  
* * * * * * * * *  
* * * * * * * * * *  
* * * * * * * * * * *  
* * * * * * * * * * * *  
* * * * * * * * * * * * *  
* * * * * * * * * * * * * *  
* * * * * * * * * * * * * * *  
* * * * * * * * * * * * * * * *  
* * * * * * * * * * * * * * * * *  
* * * * * * * * * * * * * * * * * *  
* * * * * * * * * * * * * * * * * * *  
* * * * * * * * * * * * * * * * * * * *  
* * * * * * * * * * * * * * * * * * * * *  
  
-----  
Process exited after 4.05 seconds with re  
Press any key to continue . . .
```